Midterm Exam

CEE 260/MIE 273: Probability & Statistics in Civil Engineering

October 20, 2026

time limit: Two Hours

Instructions

This exam contains **12 pages** (including the front and back pages) and **10 problems, 68 points**. You have **2 hours** to complete it. Either print out the PDF and complete by hand and upload, or complete it digitally and upload as a PDF on Canvas. This is an **open resource examination**. You are expected to complete the exam individually. Asking anyone (colleague, friend, tutor, etc) questions about the exam is **not allowed**. If any questions arise during the exam, direct them to me (via email).

The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way.
- **Show ALL your work where appropriate**. The work you show will be evaluated as well as your final answer. Thus, provide ample justification for each step you take. Indicate when you have used Python to obtain a result and show the function or statement you used to arrive at your result. In the long response questions, simply putting down an answer without showing your steps will not merit full credit. **EXCEPTION:** For short response or "True/False" questions, *no explanations are required*. However, the more work you show, the greater your chance of receiving partial credit.
- As much as possible, put your answers in the allotted space in order to facilitate grading.
- Questions are roughly in order of the lectures, so later questions may not necessarily be harder. If you are stuck on a problem, it may be better to skip it and get to it later.
- Manage your time wisely.

Problem 1 *True/False questions (10 points)*

Respond "T" (*True*) or "F" (*False*) to the following statements. Use the boxes provided. Each response is worth 1 point. Note that a statement can only be regarded as true in this framework if it always holds in all circumstances. If a statement does not hold under a given condition not already explicitly excluded, then it should be regarded as false.

(i) **F** The probability of an uncertain event is 0. An uncertain event has $0 < P < 1$; $P = 0$ means impossible.

(ii) **T** The area under the curve of a PDF is equal to 1.

(iii) **F** If the mode of a distribution is equal to the mean, then the distribution is not symmetric. E.g. the normal distribution has mode = mean and is symmetric.

(iv) **F** $P(A \cap B) = P(A) + P(B)$ for two events A and B that are mutually exclusive. $P(A \cap B) = 0$; it is $P(A \cup B)$ that equals $P(A) + P(B)$.

(v) **T** If A and B are statistically independent events, then $P(A|B) = P(A)$.

(vi) **F** For a given online account, your case-sensitive password must have 10 alphanumeric characters. Assuming you are the first user to create an account, the number of possibilities for your password is 36^{10}. Case-sensitive: $26 + 26 + 10 = 62$ characters, so 62^{10}.

(vii) **T** Under certain conditions, a binomial distribution with parameters (n, p) can be approximated by a normal distribution with $\mu = np$.

(viii) **T** Given a normal distribution with parameters μ and σ, the median of the distribution is μ.

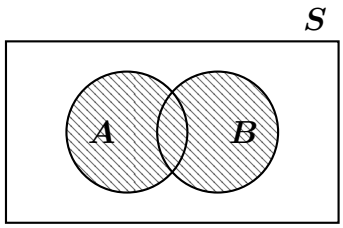
(ix) **F** In the Poisson distribution, the mean and standard deviation are equal. The mean and *variance* are equal ($= \lambda$).

(x) **T** The exponential distribution can be used to model the elapsed time between two successive events in a Poisson process.

Problem 2 *Venn diagrams (6 points)*

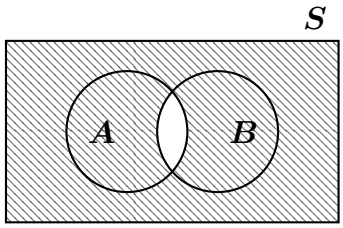
Write the combination of events (using set notation) depicted in each of the figures below.

(a)



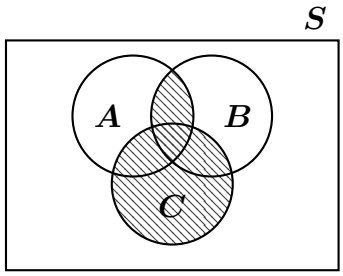
$A \cup B$

(b)



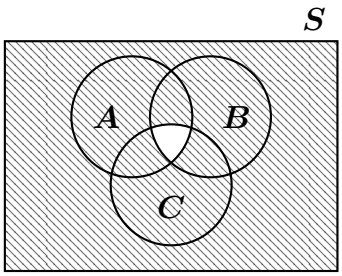
$(A \cap B)^c$ or $\overline{AB}$

(c)



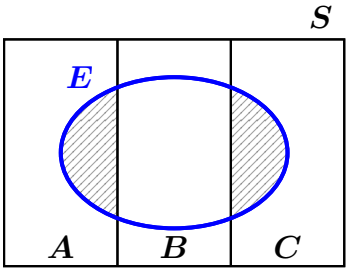
$AB \cup C$

(d)



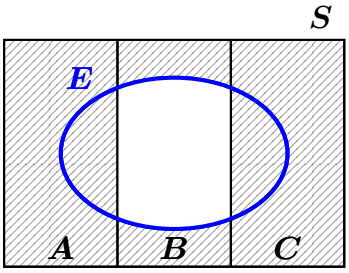
$(ABC)^c$ or $\overline{ABC}$

(e)



$EA \cup EC$ or $E\overline{B}$

(f)



$(EB)^c$ or $\overline{EB}$

Problem 3 *Counting principles (8 points)*

[2] **(a)** How many ways can 10 books be arranged on a shelf if 3 of the books are identical?

Permutations of 10 items, dividing out the 3! orderings of the identical books:

$$\frac{10!}{3!} = \frac{3\,628\,800}{6} = 604\,800$$

[6] **(b)** In a class of 170 students, if 140 are single majors and 30 are double majors, how many possible committees consisting of 4 single majors and 2 double majors can be formed? Show all your steps for maximum credit.

Order does not matter, so we use combinations for each group and multiply (fundamental counting principle):

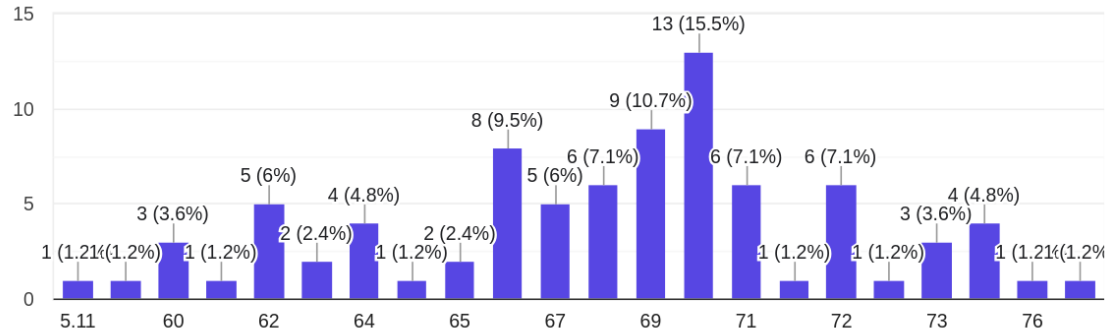
$$\begin{aligned}
\text{Single majors} &= \binom{140}{4} = \frac{140 \times 139 \times 138 \times 137}{4!} = 15\,329\,615 \\
\text{Double majors} &= \binom{30}{2} = \frac{30 \times 29}{2!} = 435 \\
\text{Committees} &= \binom{140}{4}\binom{30}{2} = 15\,329\,615 \times 435 \\
&= 6\,668\,382\,525 \approx 6.67 \times 10^9
\end{aligned}$$

In Python: `math.comb(140, 4) * math.comb(30, 2)`

Problem 4 *Basic statistics from data (3 points)*

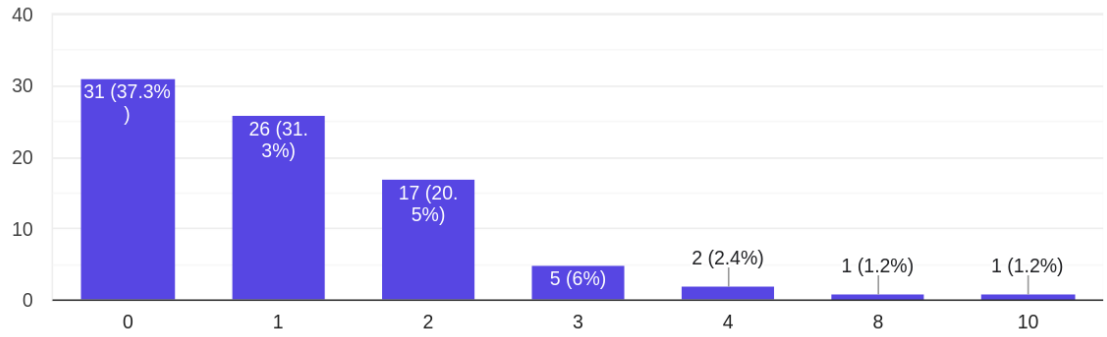
86 of you completed the Data About Us survey at the beginning of this course. Use the figures shown below to answer the following questions.

(a) What is the modal height of students in the class based on this sample ? *[1]*



70 inches

(b) What is the modal number of pets owned by each student in the class based on this sample? *[1]*



0

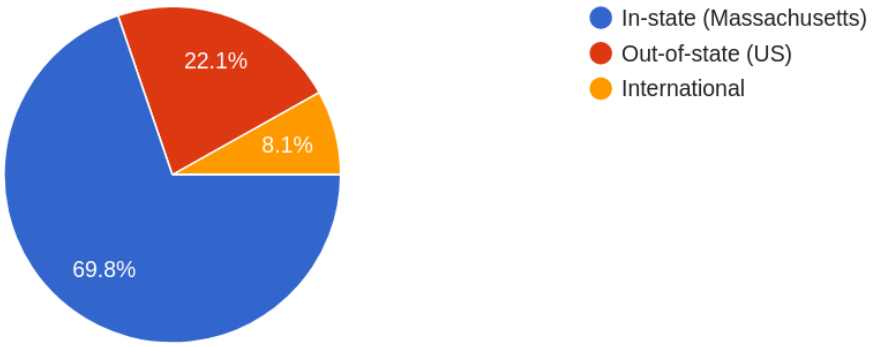
(c) Is the distribution of the number of pets owned by students in this class symmetric, left-skewed, or right-skewed? *[1]*

Right-skewed

Problem 5 *Probabilities from data (4 points)*

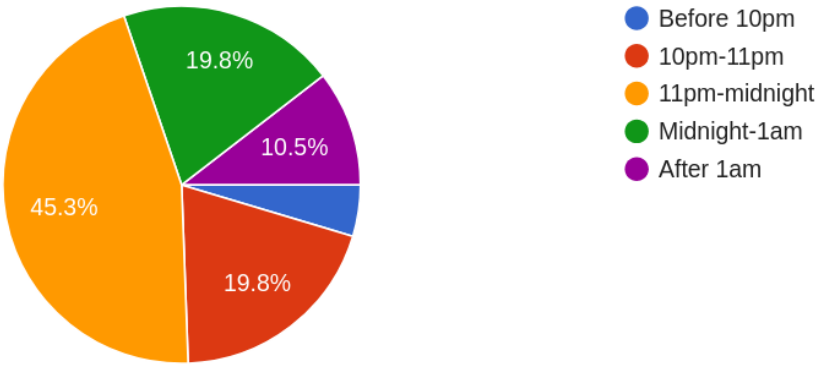
86 of you completed the Data About Us survey at the beginning of this course. Use the figures shown below to answer the following questions. Show as much work as possible and write your final answer as a decimal rounded to 2 decimal places.

[2] **(a)** What is the probability that a randomly selected student is either from out of state (i.e. not from Massachusetts) *or* international?



$$P(\text{Out-of-state} \cup \text{Intl}) = 0.221 + 0.081 \approx 0.30$$

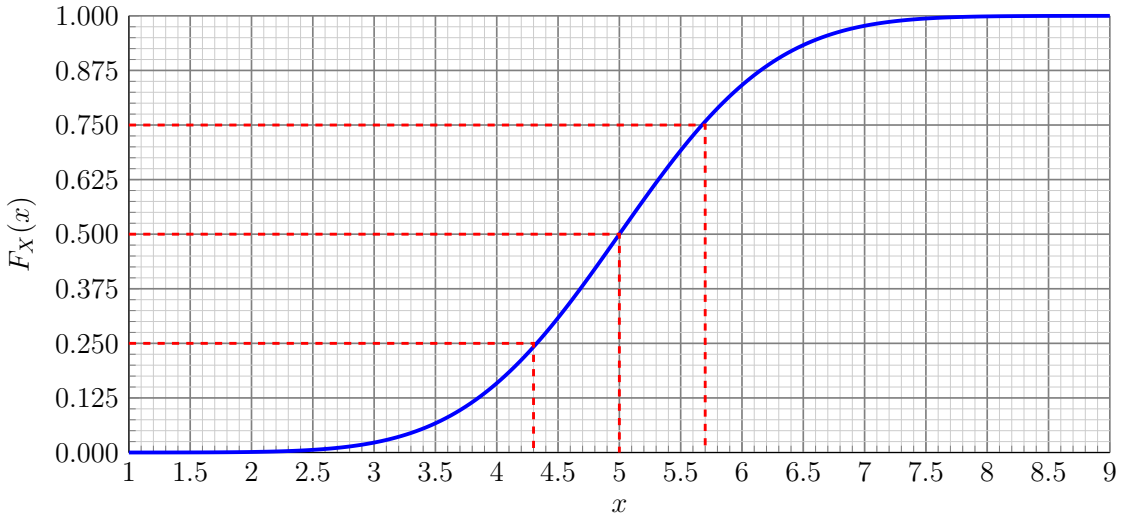
[2] **(b)** What is the probability that a randomly selected student goes to bed *before* midnight?

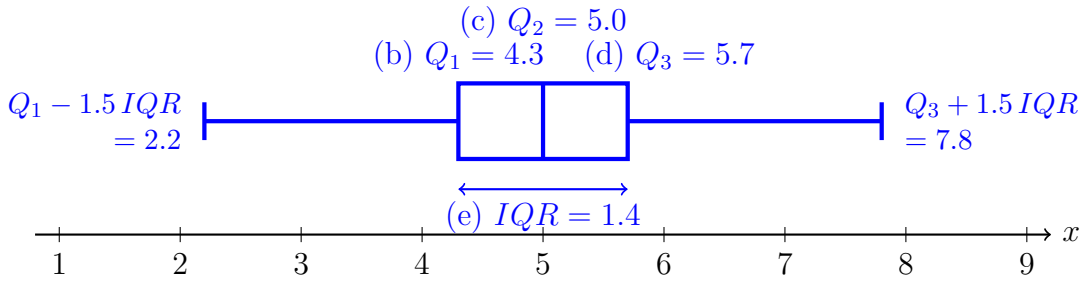


$$1 - (0.198 + 0.105) = 0.697 \approx 0.70$$

Problem 6 *CDF and boxplot (5 points)*

Use the CDF given below to sketch the corresponding boxplot of the distribution. Along with one point for the boxplot itself (a), further points will be awarded for correctly labeling: (b) the first quartile, (c) the median, (d) the third quartile, and (e) the interquartile range. (All numbers should be rounded to 1 decimal place.)





Reading off the CDF at $F_X(x) = 0.25, 0.5, 0.75$: $Q_1 \approx 4.3$, $Q_2 = 5.0$, $Q_3 \approx 5.7$, so $IQR = Q_3 - Q_1 = 5.7 - 4.3 = 1.4$. Whiskers are drawn to $Q_1 - 1.5\,IQR = 2.2$ and $Q_3 + 1.5\,IQR = 7.8$.

Problem 7 *Probabilities (10 points)*

The table below summarizes the results of a survey we did in class. Answer the following questions. Final probabilities should be given as decimals (2 decimal places). Show as much work as possible.

		What type of laptop do you use?			
		MacBook	**Windows**	**Other**	**Grand Total**
What is the type/OS of your cellphone?	**Android**	0	5	0	5
	iPhone	27	96	9	132
Grand Total		**27**	**101**	**9**	**137**

[2] **(a)** What is the probability that a randomly selected student uses an iPhone?

$P(\text{iPhone}) = \dfrac{132}{137} \approx 0.96$

[2] **(b)** What is the probability that a randomly selected student uses both an iPhone and a MacBook?

$P(\text{iPhone} \cap \text{MacBook}) = \dfrac{27}{137} \approx 0.20$

[3] **(c)** What is the probability that a randomly selected student uses an iPhone given that they use a Windows laptop?

$$P(\text{iPhone}|\text{Windows}) = \frac{P(\text{iPhone} \cap \text{Windows})}{P(\text{Windows})} = \frac{\frac{96}{137}}{\frac{101}{137}} = \frac{96}{101} \approx 0.95$$

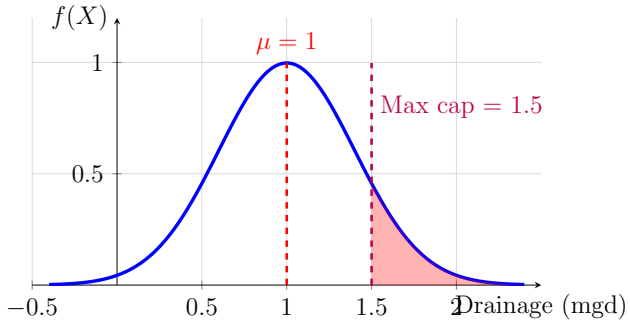
[3] **(d)** If you round probabilities to 1 decimal place, are the events "uses an iPhone" and "uses a Windows laptop" independent?

Yes. To 1 decimal place, $P(\text{iPhone}|\text{Windows}) = 0.95 \approx 1.0$ and $P(\text{iPhone}) = 0.96 \approx 1.0$, so $P(\text{iPhone}|\text{Windows}) = P(\text{iPhone})$.
Equivalently, $P(\text{iPhone} \cap \text{Windows}) = \frac{96}{137} \approx 0.7$ and $P(\text{iPhone})P(\text{Windows}) = \frac{132}{137} \cdot \frac{101}{137} \approx 0.71 \approx 0.7$.

Problem 8 *Normal distribution (9 points)*

The drainage from a community during a storm is a normal random variable estimated to have a mean of 1 million gallons per day (mgd) and an SD of 0.4 mgd. If the storm drain system is designed with a maximum drainage capacity of 1.5 mgd, answer the following questions (and show the code you use where appropriate).

(a) Sketch the normal distribution, showing both the mean and the maximum drainage capacity. *[3]*



(b) What is the underlying probability of flooding during a storm that is assumed in the design of the drainage system? (Note: flooding occurs when the maximum drainage capacity is exceeded.) *[2]*

Using the `norm.cdf()` function from the `scipy.stats` library in Python:
$P(X > 1.5) = 1 - P(X \leq 1.5) = 1 - \texttt{norm.cdf}(1.5, 1, 0.4) = 1 - \Phi(1.25) \approx 0.11$
or 11%

(c) What is the probability that the drainage is less than 1 mgd? *[2]*

Since 1 mgd is the mean of a symmetric distribution:
$P(X < 1) = \texttt{norm.cdf}(1, 1, 0.4) = \Phi(0) = 0.5$ or 50%

(d) Find the 90th-percentile drainage load from the community during a storm. *[2]*

Using the `norm.ppf()` function from the `scipy.stats` library in Python:
$x_{0.90} = \texttt{norm.ppf}(0.90, 1, 0.4) = 1 + 1.2816(0.4) \approx 1.51$ mgd

Problem 9 *Lognormal distribution (8 points)*

The distribution of the number of words (sentence length) from random sentences collected in class fitted a lognormal distribution with parameters $\mu = 2.27, \sigma = 0.67$. Now, use this distribution to answer the following questions.

[2] (a) What is the median sentence length?

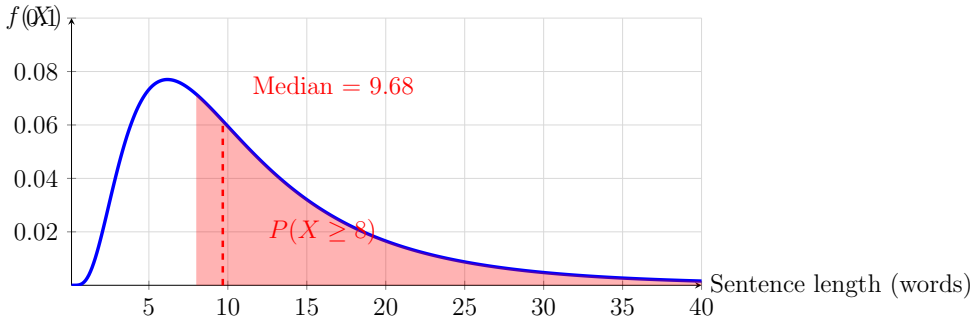
The median of a lognormal distribution is given by e^{μ}.
$Median = e^{2.27} \approx 9.68$ words

[3] (b) What is the probability that the sentence length is at least 8 words?

Using the `lognorm.cdf()` function from the `scipy.stats` library in Python:

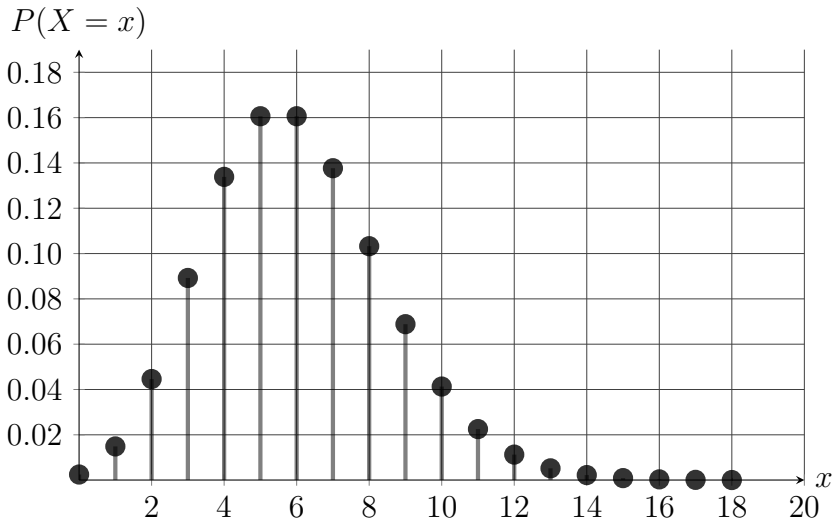
$$\begin{aligned} P(X \geq 8) &= 1 - P(X < 8) \\ &= 1 - \Phi\left(\frac{\ln 8 - 2.27}{0.67}\right) = 1 - \Phi(-0.284) \\ &= 1 - \texttt{lognorm.cdf(8, 0.67, scale=np.exp(2.27))} \\ &\approx 0.61 \text{ or } 61\% \end{aligned}$$

[3] (c) Sketch the distribution, indicating the median and shading the portion of the PDF corresponding to the probability in part (b).



Problem 10 *Discrete distribution (5 points)*

The PMF of a random variable X is given in the figure below. Use the figure to answer the following questions.



(a) Estimate the probability $P(X = 5)$. *[1]*

$$0.16$$

(b) Estimate the probability $P(6 \leq X < 10)$. *[2]*

$$= \sum_{x=6}^{9} P(X = x)$$
$$= 0.16 + 0.14 + 0.10 + 0.07 = 0.47$$

(c) If the distribution shown has a single parameter λ, what distribution is it? *[1]*

Poisson distribution

(d) Which of the following would be your best guess for λ? (Circle or underline the correct answer.) *[1]*

i. 4 **ii.** (6) **iii.** 8 **iv.** 10



CEE 260/MIE 273 | Jimi Oke | Fall 2026
Department of Civil and Environmental Engineering
University of Massachusetts Amherst